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Institut de chimie du Canada | **Pour notre avenir**

CANADIAN CHEMISTRY OLYMPIAD | OLYMPIADE DE CHIMIE DU CANADA

THE CANADIAN CHEMISTRY CONTEST 2025
PART A – MULTIPLE CHOICE QUESTIONS (60 minutes)
All contestants should attempt this part of the contest before proceeding to Part B and/or Part C.
The only reference material allowed is the CIC/CCO Periodic Table provided.
Students may use a scientific calculator. No phones or communication devices are allowed.

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1) Which of the following reactions should be conducted in a fume hood?

I. Placing 20 g of solid Mn in 50 mL of water
II. Mixing 25 mL 0.10 mol L^{-1} HCl and 50 mL 0.05 mol L^{-1} NaOH
III. Placing 2.0 g of MnO_2 in 30% H_2O_2

A) I only B) II only C) III only D) I and II E) II and III

2) Which of the following elements would have the highest first ionization energy?

A) Phosphorus B) Sulfur C) Lithium D) Chromium E) Sodium

3) Which of the following would cause largest change in the strength of the intermolecular forces?

A) Dissolving NaCl in liquid ammonia instead of water
B) Substituting one H atom in CH_4 with a F atom to form CH_3F
C) Increasing the temperature of liquid water from 20°C to 80°C
D) Replacing the -OH in ethanol with a-CH_3 group to form propane
E) Transforming diamond into graphite

4) For the chlorite ion ClO_2^-, which of the following is closest to the expected O-Cl-O bond angle?

A) 180° B) 120° C) 110° D) 90° E) 45°

5) If 22.3 g of an elemental gas occupies 20.5 L at 49.2 ˚C and 780 torr, what is the most likely identity of the gas? Assume ideal gas behavior.

A) N_2 B) Cl_2 C) O_2 D) Xe E) Kr

6) Phosphoric acid can be produced by the reaction of tetraphosphorus decoxide with water, according to the **unbalanced** equation below:

$$P_4O_{10\ (s)} + H_2O_{\ (\ell)} \rightarrow H_3PO_{4\ (aq)}$$

What mass of P_4O_{10} would need to be added to 2.00 L of water to produce a 0.0200 mol L^{-1} solution of phosphoric acid? Assume 100% yield in this reaction.

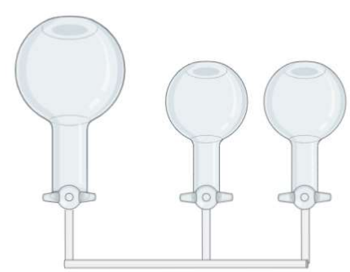
A) 1.14 g B) 5.68 g C) 2.28 g D) 2.84 g E) 11.4 g

7) In the periodic table, metals are on the left side and non-metals are on the right. Which of the following is **NOT true**?

A) As atomic radius increases, so does metallic character
B) As effective nuclear charge increases, so does metallic character
C) As electron shielding increases, so does metallic character
D) As electronegativity increases, metallic character decreases
E) As ionization energy decreases, metallic character increase.

8) Consider the following apparatus, where each of the bulbs contains a gas at the pressure shown. If the temperature remains constant and the volume of the tubing connecting the bulbs is negligible, what is the pressure inside the apparatus when all of the valves are open?

1.0 L N_2	0.5 L Ne	0.5 L He
255 torr	700 torr	465 torr



A) 419 torr
B) 700 torr
C) 1420 torr
D) 710 torr
E) 350 torr

9) Which of the following elements forms a stable +3 ion with the shorthand electron configuration: [noble gas]*(n-1)*d^2?

A) Zr B) Tc C) Sn D) Mo E) V

10) A pure gold coin is heated to 80.0 °C and placed in 500.0 mL of water chilled to 5.00°C. The mixture of gold and water reaches a maximum temperature of 6.36 °C. If the density of water is 1.00 g mL^{-1}, the specific heat capacity of water is 4.184 J g^{-1} $°C^{-1}$ and the specific heat capacity of gold is 0.129 J g^{-1} $°C^{-1}$, what is the mass of the gold coin?

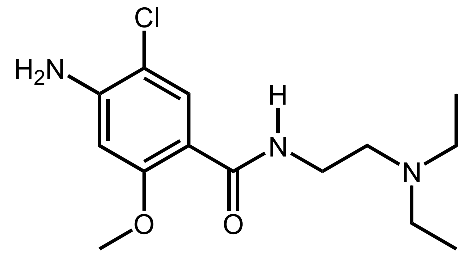
A) 26.6 g B) 9.50 g C) 300 g D) 28.6 g E) 2.85 g

11) A dynamic density bottle contains two liquid layers: a solution of saltwater (density 1.12 g mL^{-1}) and isopropanol (density 0.79 g mL^{-1}). When the bottle is vigorously shaken, the two layers form a temporary emulsion layer (density 0.80 – 1.10 g mL^{-1}). The bottle also contains small pellets made of two plastics, polyethylene (density 0.92-0.94 g mL^{-1}) and polystyrene (density 1.05 – 1.07 g mL^{-1}).

Before the bottle is shaken, which of the following scenarios would most likely result in the polystyrene pieces floating at the interface between the isopropanol and the saltwater solution layers rather than sinking into the saltwater solution layer?

A) Increasing the salt concentration in the saltwater solution.
B) Replacing isopropanol with a less dense alcohol, such as ethanol.
C) Adding a small amount of surfactant to the saltwater to reduce the surface tension.
D) Decreasing the temperature of the bottle and its contents.
E) Increasing the proportion of isopropanol in the mixture.

Questions 12 - 14 involve molecule A below:



A

12) Which of the following functional groups are in the molecule labelled “A”.

A) aliphatic amine, ketone, ether
B) amide, ether, aromatic halide
C) aromatic amine, ketone, ether
D) aromatic halide, aromatic amine, ketone
E) amide, ether, aliphatic halide

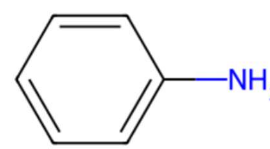
13) If an analytical chemist conducted the accurate elemental analysis of compound A what would be the percentage of carbon?

A) 60.01 % B) 59.51% C) 56.09 %
D) 55.23 % E) 54.78%

14) How many of the carbon atoms in compound A are best described as sp^2-hybridized?

A) 6 B) 7 C) 8 D) 9 E) All of the carbon atoms

15) The pH of a 0.20 mol L^{-1} solution of the weak base with the following structure is 8.95. What is the pK_b of the base?



A) 3.97 B) 5.05 C) 8.95
D) 9.40 E) 10.10

16) Given the equilibrium N_2O_4 (g) $\rightleftharpoons$ 2 NO_2 (g) $\Delta H = +58$ kJ mol^{-1}

Which of the following conditions will shift the equilibrium to the right?

I. Increasing the temperature
II. Decreasing the pressure
III. Adding a catalyst

A) I and II only B) I and III only C) II and III only
D) I, II and III E) none

17) The rate determining step for the reaction below is 3rd order

$$2\ A_2B + 3\ C_2 \rightarrow 4\ AC + 2\ BC$$

Given the data in the table, what is the most likely rate law?

Trial	$[A_2B]$	$[C_2]$	Reaction Rate (mol L^{-1} s^{-1})
1	0.10	0.10	1.2×10^{-2}
2	0.20	0.10	2.4×10^{-2}

A) Rate = $k[A_2B]^2[C_2]^3$ B) Rate = $k[A_2B]^3$
C) Rate = $k[A_2B]$ D) Rate = $k[A_2B][C_2]^2$
E) Rate = $k[A_2B]^2[C_2]$

18) A student decides to empirically determine the heat of vaporization of water using an electric kettle and records the following data:
Initial water temperature: 20 °C
Mass of water: 500 g
Time to reach boiling point (100°C): 2 min
Mass of water evaporated in 1min at the boiling temperature: 50 g
Specific heat capacity of water: 4.184 J g^{-1} $°C^{-1}$

Given the student's data, what is the best estimate of the heat of vaporization of water?

A) 30.2 kJ mol^{-1} B) 40.7 kJ mol^{-1} C) 54.0 kJ mol^{-1}
D) 75.3 kJ mol^{-1} E) 60.2 kJ mol^{-1}

19) When the following equation is balanced in presence of water, what is the SUM of the coefficients?

NO_2^- (aq) + Al (s) $\rightarrow$ NH_4^+ (aq) + AlO_2^- (aq)

A) 8 B) 10 C) 6 D) 7 E) 5

20) Silver sulfate, Ag_2SO_4, has a $K_{sp} = 1.4 \times 10^{-4}$. How many silver ions would be dissolved in 25.0 mL of a saturated solution of silver sulfate?

A) 4.9×10^{20} B) 7.8×10^{20} C) 9.8×10^{20}
D) 1.2×10^{21} E) 2.0×10^{22}

21) Stereoisomers of organic compounds have the same connectivity of atoms but different spatial arrangements. Conformation isomers are stereoisomers that are related by rotations about single bonds. **Ignoring conformational isomers**, how many stereoisomeric forms are there for the molecule 5-bromo-1-pentene?

A) 0 B) 2 C) 4 D) 16 E) 17

22) Assuming standard conditions, arrange the following in order of decreasing strength as reducing agents in acidic solution: Fe, Br^-, Sn^{2+}, H_2O_2, Mg

$Fe^{2+} + 2e^- \rightleftharpoons Fe$	$E° = -0.45$ V
$2Br^- \rightleftharpoons Br_2(l) + 2e^-$	$E° = -1.09$ V
$Sn^{2+} \rightleftharpoons Sn^{4+} + 2e^-$	$E° = -0.15$ V
$O_2 + 2H^+ + 2e^- \rightleftharpoons H_2O_2$	$E° = 0.70$ V
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	$E° = -2.37$ V

A) Mg, Fe, Sn^{2+}, H_2O_2, Br^- B) Mg, H_2O_2, Sn^{2+}, Fe, Br^-
C) Br^-, Fe, Sn^{2+}, H_2O_2, Mg D) H_2O_2, Sn^{2+}, Fe, Br^-, Mg
E) Mg, Br^-, Fe, Sn^{2+}, H_2O_2

23) Many common bleaching and disinfecting agents are dilute solutions of sodium hypochlorite (NaOCl). If the K_a of hypochlorous acid (HOCl) at SATP is 3.0×10^{-8}, what is the pH of a 0.10 mol L^{-1} solution of sodium hypochlorite?

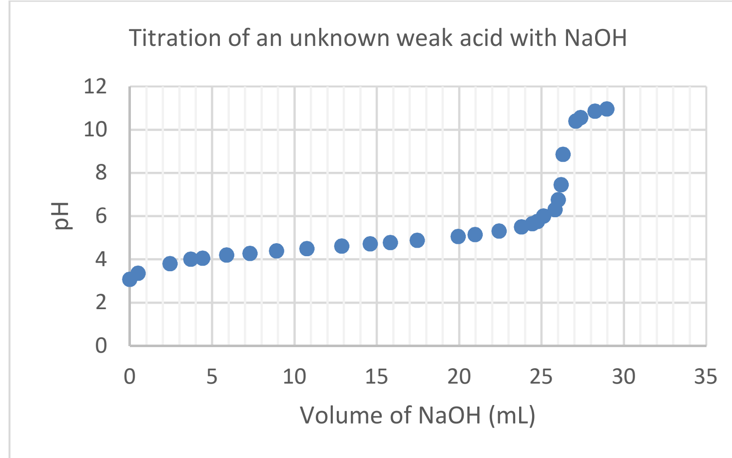
A) 3.74 B) 6.48 C) 7.52 D) 8.63 E) 10.26

24) The balanced thermochemical reaction for the complete combustion of oxalic acid with the lowest whole number coefficients is
$2\ H_2C_2O_4\ (s) + O_2\ (g) \rightarrow 4CO_2\ (g) + 2H_2O\ (g)$ $\Delta H° = -492.1$ kJ
Given the following thermodynamic data, determine the $\Delta G°$ for the complete combustion of 1 mol of oxalic acid.

Substance	$\Delta H_f°$ (kJ mol^{-1})	S° (J mol^{-1} K^{-1})
C (s)	0	5.69
CO_2 (g)	–393.5	213.6
H_2 (g)	0	130.6
H_2O (g)	–241.8	188.7
O_2 (g)	0	205.0
$H_2C_2O_4$ (s)	?	120.1

A) –782.8 kJ mol^{-1} B) –363.3 kJ mol^{-1} C) 393.3 kJ mol^{-1}
D) – 900.0 kJ mol^{-1} E) 1565.5 kJ mol^{-1}

25) During a titration analysis, the concentration of a known titrant is often determined using a primary standard. For the titration of an unknown weak acid with NaOH, the primary standard, potassium hydrogen phthalate was used. The standardized concentration of the NaOH solution was determined to be 0.101 mol L^{-1}. A volume of 10.0 mL of an unknown monoprotic weak acid was titrated with the standardized NaOH and the following data was collected What is the best estimate of the K_a of the unknown monoprotic weak acid?



Volume NaOH (mL)	pH	Volume NaOH (mL)	pH
0.00	2.66	20.98	5.15
0.50	3.35	22.43	5.31
2.45	3.80	23.78	5.50
3.70	4.00	24.45	5.65
4.42	4.05	24.77	5.75
5.88	4.20	25.13	6.00
7.31	4.28	25.82	6.30
8.91	4.39	26.03	6.75
10.75	4.49	26.20	7.45
12.88	4.62	26.32	8.85
14.61	4.72	27.10	10.40
15.83	4.78	27.38	10.55
17.45	4.88	28.25	10.85
19.95	5.05	28.98	10.95

A) 4.96×10^{-10} B) 7.08×10^{-6} C) 1.41×10^{-9}
D) 2.02×10^{-5} E) 5.01×10^{-11}

End of Part A of the contest
Go back and check your work

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Data Sheet
Fiche de données

CHEMISTRY **Olympiads**
CANADA
Olympiades CHIMIE

Relative Atomic Masses (2012, IUPAC)
***For the radioactive elements the atomic mass of an important isotope is given**

Masses Atomiques Relatives (UICPA, 2012)
***Dans le cas des éléments radioactifs, la masse atomique fournie est celle d'un isotope important**

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 **H** 1.008																	2 **He** 4.003
3 **Li** 6.941	4 **Be** 9.012											5 **B** 10.81	6 **C** 12.01	7 **N** 14.01	8 **O** 16.00	9 **F** 19.00	10 **Ne** 20.18
11 **Na** 22.99	12 **Mg** 24.31											13 **Al** 26.98	14 **Si** 28.09	15 **P** 30.97	16 **S** 32.07	17 **Cl** 35.45	18 **Ar** 39.95
19 **K** 39.10	20 **Ca** 40.08	21 **Sc** 44.96	22 **Ti** 47.87	23 **V** 50.94	24 **Cr** 52.00	25 **Mn** 54.94	26 **Fe** 55.85	27 **Co** 58.93	28 **Ni** 58.69	29 **Cu** 63.55	30 **Zn** 65.38	31 **Ga** 69.72	32 **Ge** 72.61	33 **As** 74.92	34 **Se** 78.96	35 **Br** 79.90	36 **Kr** 83.80
37 **Rb** 85.47	38 **Sr** 87.62	39 **Y** 88.91	40 **Zr** 91.22	41 **Nb** 92.91	42 **Mo** 95.96	43 **Tc** (98)	44 **Ru** 101.1	45 **Rh** 102.9	46 **Pd** 106.4	47 **Ag** 107.9	48 **Cd** 112.4	49 **In** 114.8	50 **Sn** 118.7	51 **Sb** 121.8	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.3
55 **Cs** 132.9	56 **Ba** 137.3	57 **La** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.9	75 **Re** 186.2	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209.0	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 **Ac** (227)	104 **Rf** (261)	105 **Db** (262)	106 **Sg** (266)	107 **Bh** (264)	108 **Hs** (277)	109 **Mt** (268)	110 **Ds** (269)	111 **Rg** (272)	112 **Cn** (285)	113 **Nh** (284)	114 **Fl** (289)	115 **Mc** (288)	116 **Lv** (292)	117 **Ts** (294)	118 **Og** (294)

58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3	65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
90 **Th** 232.0	91 **Pa** (231.0	92 **U** (238.0)	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (262)

	Symbol / Symbole	**Value / Quantité numérique**	
Atomic mass unit	*amu*	1.66054×10^{-27} kg	Unité de masse atomique
Avogadro's number	N_A	6.022×10^{23}	Nombre d'Avogadro
Charge of an electron	e	1.60218×10^{-19} C	Charge d'un électron
Dissociation constant (H_2O)	K_w	1.00×10^{-14} (25ºC)	Constante de dissociation de l'eau (H_2O)
Faraday's constant	F	96 485 C mol^{-1}	Constante de Faraday
Gas constant	R	8.31451 J K^{-1} mol^{-1}	Constante des gaz
		0.08206 L atm K^{-1} mol^{-1}	
Mass of an electron	m_e	9.10939×10^{-31} kg	Masse d'un électron
Mass of a neutron	m_n	1.67493×10^{-27} kg	Masse d'un neutron
Mass of a proton	m_p	1.67262×10^{-27} kg	Masse d'un proton
Planck's constant	h	6.62608×10^{-34} J s	Constante de Planck
Speed of light	c	2.997925×10^{8} m s^{-1}	Vitesse de la lumière
Rydberg constant	R_H	1.096×10^{7} m^{-1}	Constante de Rydberg

1 Å	= 1×10^{-10} m
1 atm	= 101.325 kPa
1 bar	= 1×10^{5} Pa

STP/TPN	**SATP/TPAN**
273.15 K	298 K
100 kPa	100 kPa